



Toxicological safety evaluation of Qiguiyin formula in rats at the treatment phase and recovery phase

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ABSTRACT

Ethnopharmacological relevance: Qiguiyin, a hospital preparation of traditional Chinese medicinal formula, is a combination of *Astragalus hamosus* L., *Angelica sinensis* (Oliv.) Diels, *Lonicera sempervirens* L., *Artemisia annua* L., and *Polygonum cuspidatum* Siebold & Zucc. at a ratio of 12:3:3:2:2. It has been used to treat severe pneumonia caused by drug-resistant bacteria in clinical practice, while studies on its toxicological safety are rare in the literature.

Aim of the study: In the present study, we aimed to develop a new application of Qiguiyin according to the general research routine of traditional Chinese medicine (TCM), and the toxicological effects of the Qiguiyin formula at the treatment phase and recovery phase were also evaluated.

Materials and methods: The rats were administered with the Qiguiyin formula at 10, 30, and 50 times of the corresponding dosage in humans for 13 consecutive weeks. During 13 weeks of the treatment phase and 4 weeks of the recovery phase, the general signs of toxicity and mortality were monitored daily, and the body weight and food consumption were determined every week. Moreover, the hematology, biochemistry, urine, organ weights, and histopathology were analyzed, and the reproductive system was examined at the end of the treatment phase or recovery phase, respectively.

Results: The toxicological results showed no deaths and no changes in general behavior. Moreover, there was no clinically significant effect of the Qiguiyin formula on body weight or food consumption in rats. Although the Qiguiyin formula resulted in some changes in hematological, biochemical, and urinary indexes, these alterations were not related to the treatment because they remained within normal ranges throughout the 17 weeks. Besides, the main organs were not affected basically. All the above-mentioned results showed no gender difference. Furthermore, a clinical dosage of 50 times of the Qiguiyin formula did not affect the reproductive system of female rats, while it could lead to atrophied seminiferous tubules in two out of 10 male rats. However, such abnormality could not be found at the end of the recovery phase.

Conclusions: Overall, the Qiguiyin formula could be used safely. The administration at doses of less than 1000 g/day for 13 weeks showed no distinct toxicity or side effects.

1. Introduction

Herbal medicines have been widely applied for thousands of years worldwide, which are especially considered to be safe based on experience (Gao et al., 2018b; Qi et al., 2020). Many previous studies have reported the pharmacological activities of five components (at a ratio of 12:3:3:2:2) in Qiguiyin, such as *A. hamosus*, *A. sinensis*, *L. sempervirens*,

A. annua, and *P. cuspidatum* (Bahaeddin et al., 2018; Cheng et al., 2020; Nair et al., 2021).

A. hamosus, generally listed in traditional formulas including Buzhongyiqitang, Bufeitang, Yupingfengsan, and Dangguibuxuetang, is the core component of Qiguiyin. Modern studies have shown the efficacy of *A. hamosus* for strengthening the spleen and nourishing qi, and it has an immunomodulatory effect on repeated respiratory infections (Li et al., 2018a, 2019). Chemically, the important active ingredients of *A.*

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Abbreviations

Traditional Chinese Medicine	TCM	High Performance Liquid Chromatography	HPLC
Interferon	IFN	Evaporative Light-Scattering Detector	ELSD
Interleukin	IL	Sprague-Dawley	SD
Transforming Growth Factor	TGF	Specific Pathogen-Free	SPF
Retinoid-related Orphan Receptor Gamma t	ROR γ t	Control Group	CN
T-helper	Th	EDTA dipotassium salt	K2EDTA
Regulatory T	Treg	Hematoxylin and Eosin	H&E
Allergic Rhinitis	AR	Statistical Analysis System	SAS
<i>Pseudomonas aeruginosa</i>	<i>P. aeruginosa</i>	Fructus Psoraleae	FP
Astragaloside IV	As-IV	Hypothalamic-Pituitary-Adrenal	HPA
		Ethanol	EtOH
		Arecae Semen Aqueous Extract	ASAE

hamosus mainly include astragalosides, polysaccharides, and flavonoids (Gao et al., 2018a).

A. sinensis comes from Gansu Province, a traditional medicinal and edible plant, and it is included in many traditional formulas, such as Dangguibuxuetang, Siwutang, Bazhentang, and Xiaoyaosan. Its efficacy has been shown in invigorating blood, promoting blood circulation, regulating menstruation, and relieving pain in Compendium of Materia Medica. According to previous studies, it shows anti-inflammatory, antioxidant, immunomodulation, anti-arteriosclerosis, and antibacterial properties (Yang et al., 2020; Choy et al., 1994; Kim et al., 2018; Li

et al., 2020). It has been reported that *A. sinensis* can also significantly improve inflammatory cell infiltration, increase the ratios of interferon (IFN)- γ /interleukin (IL)-4 and transforming growth factor (TGF)- β /IL-17, and decrease the levels of retinoid-related orphan receptor gamma t (ROR γ t) in lung tissues. Moreover, it plays a therapeutic role in asthma with Yin deficiency syndrome, improves airway inflammation by regulating the balance of T-helper (Th) 1/Th2 and Th17/regulatory T (Treg), enriches the blood, and invigorates the circulation of blood in bronchial asthma (Ma et al., 2017; Wang et al., 2018).

L. sempervirens generally comes from Henan Province and Shandong

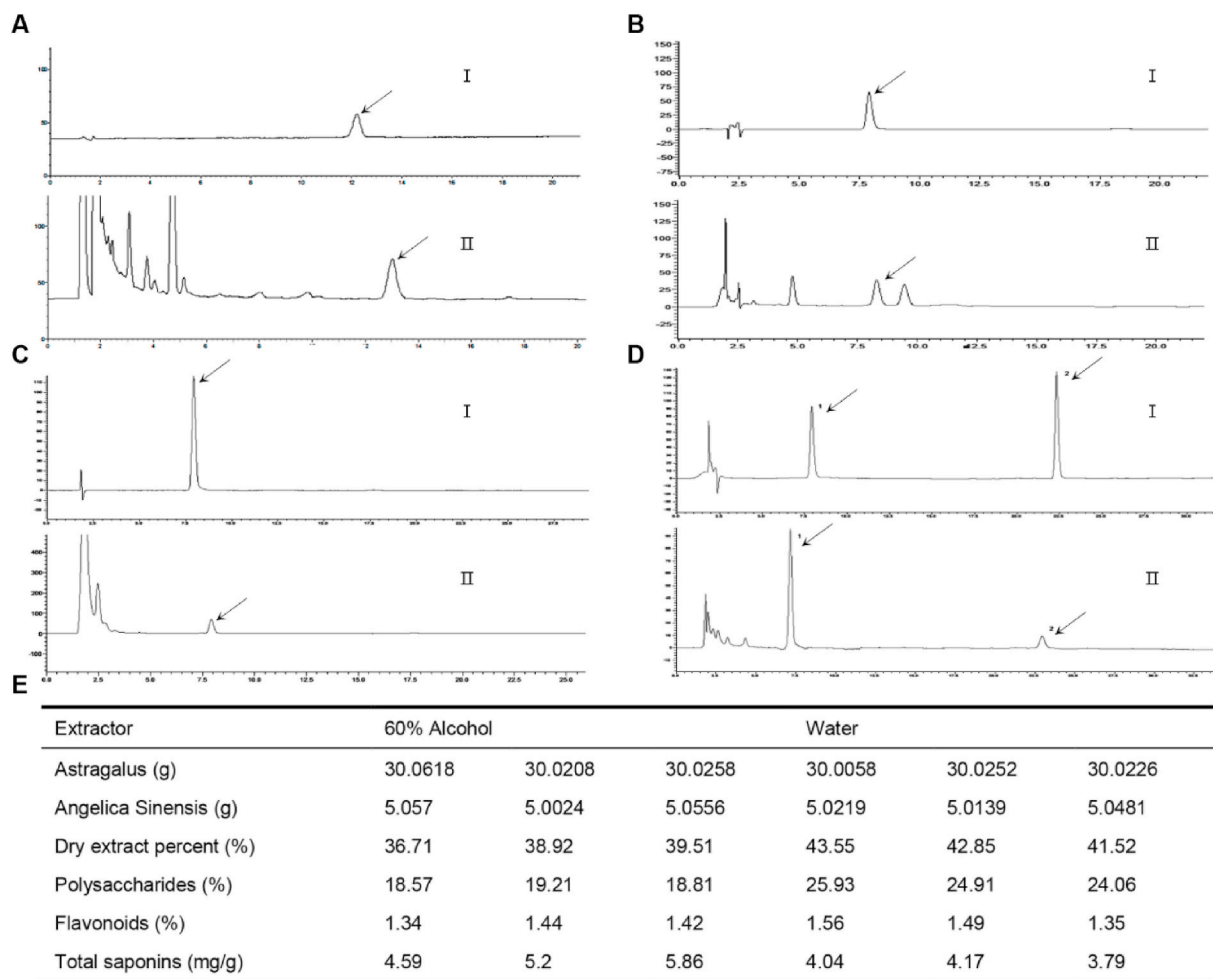


Fig. 1. Confirmation results of four vital pharmacological components of Qiguiyin. (A) Chromatogram analysis of As-IV (B) Chromatogram analysis of chlorogenic acid (C) Chromatogram analysis of emodin (D) Chromatogram analysis of polydatin (I, reference standard, II, Qiguiyin sample) by HPLC. The arrows indicate the peak. (E) Extraction quantity comparison of total saponins, flavonoids, and polysaccharides with two different solvents (60% alcohol and water).

Province, and it is an extremely used traditional edible-medicinal herb and widely included in many traditional formulas, such as Xianfan-ghuomingyin, Yinqiaosan, Qingyingtang, and Wuweixiaoduyin. Its efficacy has been shown in evacuating wind-heat, clearing heat, and detoxifying in Compendium of Materia Medica. In recent modern pharmacological studies, *L. sempervirens* shows antiviral, antioxidative, anti-inflammatory properties, and immune-regulatory functions (Su et al., 2021), while its extraction can effectively relieve ovalbumin-induced allergic rhinitis (AR) (Lin et al., 2019). Chlorogenic acid is an important material basis for *L. sempervirens* to play its biological function.

A. annua and *P. cuspidatum* are also indispensable components of the Qiguiyin formula. According to the TCM prescription principle, they both play important roles in the Qiguiyin formula. Especially, emodin and polydatin are important active ingredients of *P. cuspidatum*.

Our previous research has also shown the efficacy of Qiguiyin on immunomodulation and nourishing qi in pneumonia (Ding et al., 2019; Kong et al., 2015). Besides, serum from Qiguiyin-treated rats can effectively inhibit antibiotic resistance of *Pseudomonas aeruginosa* (*P. aeruginosa*) (Ding et al., 2020). However, only very few studies have investigated its toxicological safety. Therefore, there is a lack of modern scientific evidence regarding the safety of Qiguiyin.

Since Astragaloside IV (As-IV), chlorogenic acid, emodin, and polydatin are the main pharmacological ingredients of Qiguiyin, in the present study, we aimed to evaluate the subchronic toxicity of Qiguiyin. Rats were administered with 10, 30, and 50 times of human doses of Qiguiyin for 13 weeks, followed by 4 weeks of the recovery phase, and all the relative parameters were monitored during these 17 weeks. Collectively, our findings laid a scientific foundation for the application of Qiguiyin in clinical therapy and offered valuable insights into the secondary development of Qiguiyin.

2. Materials and methods

2.1. Components and preparation of Qiguiyin

Qiguiyin was supplied by the Beijing University of Traditional Chinese Medicine (Lot No. 216–002). It was composed of *A. hamosus*, *A. sinensis*, *L. sempervirens*, *A. annua*, and *P. cuspidatum* at a ratio of 12:3:3:2:2. The five components were purchased from Beijing Xinglin Pharmaceutical Co., Ltd., which met the requirements of the Chinese Pharmacopoeia (2010 edition). Following the orthogonal design, the methods of water extraction and alcohol precipitation were used to prepare Qiguiyin. The processes of Qiguiyin preparation were as follows. Briefly, the herbs with the above-mentioned prescription ratio were decocted. The decoctions were concentrated to 1.0 g/mL. Subsequently, 12 vol of water were added to the concentrated Qiguiyin, followed by extraction for 1 h, and such procedure was repeated three times. The extraction was precipitated with 60% ethanol. The harvested Qiguiyin extraction was stored at -20°C , it was dissolved in sterile water before gavage.

The reference standards As-IV (batch number 110781–201314), chlorogenic acid (batch number 110753–201314), emodin (batch number 1110756–200110), and polydatin (batch number 111575–200502) were purchased from the China Institute of Food and Drug Control. The four above-mentioned ingredients of Qiguiyin were analyzed via Agilent 1100 High Performance Liquid Chromatography (HPLC) system using an SB-C18 column (4.6 mm $\times$ 150 mm, 4.6 μm) equipped with an evaporative light-scattering detector (ELSD) 2000 and an ultraviolet–visible light detector. The contents of the four Qiguiyin ingredients were expressed as $\mu\text{g/mL}$. As-IV, chlorogenic acid, emodin, and polydatin were confirmed as the four main active ingredients of the Qiguiyin formula by optimized HPLC (Fig. 1).

2.2. Animals and ethics

A total of 120 Sprague-Dawley (SD) rats of both sexes (specific pathogen-free (SPF) grade) (6 weeks old and weighing 155.9–174.1 g in males, and 165.6–190.9 g in females) were purchased from Beijing WeitongLihua Laboratory Animal Technology Co., Ltd. All rats were housed under standard environmental conditions ($25 \pm 1^{\circ}\text{C}$, humidity of 50%–65%, and a 12 h light/dark cycle), and free access to food and water was given to the animals. Rats were allowed to adapt to the environment for 7 days.

The animal-related protocols were approved by the Experimental Animal Ethics Committee of Beijing Hospital of Traditional Chinese Medicine (No. 2017–1007; Approval date: October 10, 2017).

2.3. Equipment and reagents

ABX-PENTRA-80 automated hematology analyzer was purchased from Germany. C2000-4 blood coagulator analyzer was obtained from Precil Ltd., Beijing China. BT-2000PLUS analyzer was supplied by Biotechnica Chemistry Co., Italy. DSI 905 electrolyte analyzer was provided by Shanghai Xunda Medical Instrument Co., Ltd.

Qiguiyin was supplied by the College of Chinese Medicine, Beijing University of Traditional Chinese Medicine (Lot No. 216–002).

2.4. Drug administration

The animals were randomly divided into a control group (CN) and three experimental groups (TA, TB, and TC) according to the Qiguiyin dosage. There were 15 female and 15 male rats in each group.

Rats in the TA, TB, and TC groups were administered with the Qiguiyin extraction at doses of 2.86 g/kg, 8.58 g/kg, and 14.30 g/kg every day, namely low-, middle-, and high-dose groups, respectively. The concentrate was 1.0 ml/100 g body weight in the TA group, but 1.3 in the TB and TC group. These above-mentioned dosages corresponded to 10, 30, and 50 times of human dosage in clinical practice. The clinical dosage of Qiguiyin extraction in humans is 600 g/day, which is corresponding to the middle dose in rats. The control group received the same volume of double-distilled water. Before the treatment, the body weight of rats was 185.6–226.8 g in males and 186.0–218.7 g in females, and a body weight of 200 g was regarded as an average value. All rats underwent gavage administration for 13 weeks (about 3 months, named treatment phase). It was worth noting that the Qiguiyin extraction at a dose of 0.286 g/kg was about the effective dose in SD rats to induce pneumonia based on the body surface area of rats and the clinical dose of humans, and the corresponding dosage in humans is 20 g/day.

$$Y = K(X \div 70)$$

Y: Corresponding dosage in rat; K: 0.2; X: Clinical dosage in human (Standard human body weight 70 kg, Clinical dosage in human 20 g/d)

At the end of the treatment phase, 10 male and 10 female rats from each group were sacrificed for the analyses of hematology, biochemistry, urine, main organ weights, and histopathology of reproductive organs. The rest rats in each group, including five females and five males, were given free access to food and water for another 4 weeks (14–17 weeks, named recovery phase), during which the treatment was terminated. At the end of the recovery phase, they were sacrificed, and all the above-mentioned parameters were assessed.

2.5. General toxicological evaluation

During the 17 weeks of the experiment, the detailed clinical signs of toxicity, including the time of onset, intensity, duration of toxicological symptoms, and mortality were monitored and recorded every day,

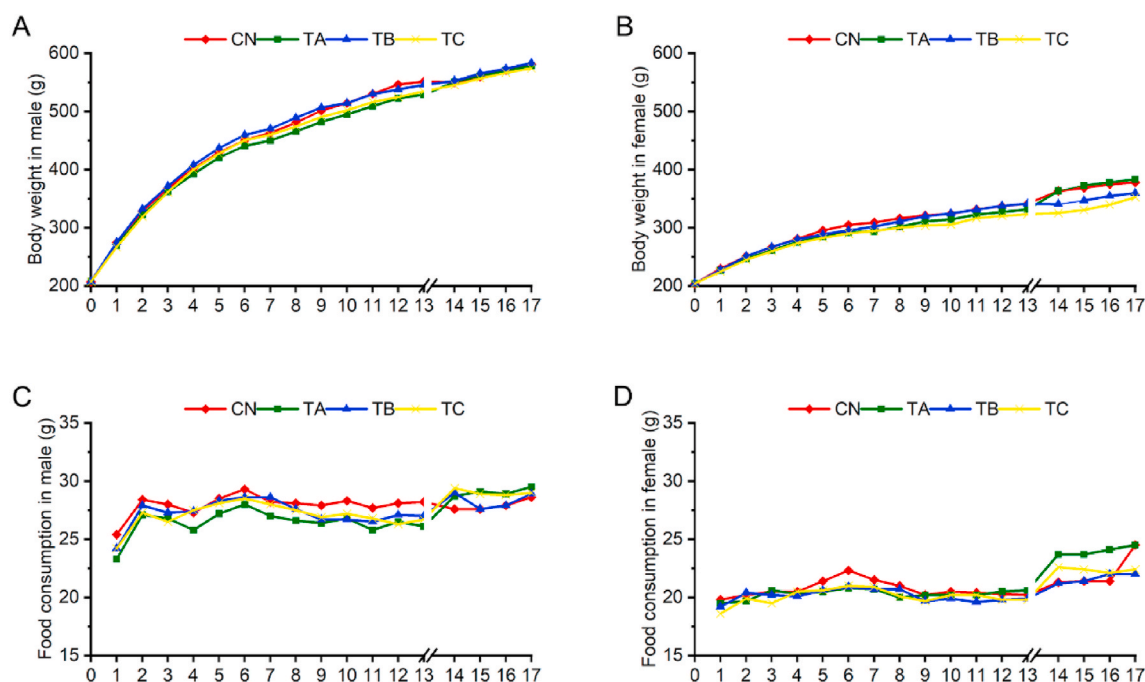


Fig. 2. Effects of different doses of Qiguiyin on the body weight in male (A) or female (B) rats and food consumption in male (C) or female (D) rats. In the treatment phase, $n = 20$ in each group, including 10 females and 10 males. In the recovery phase, $n = 10$ in each group, including five females and five males. Color images are available online. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

including changes in skin and fur, eyes and mucous membranes, manure, psyche states, and behavioral patterns. Especially, the number of mortality was noted.

2.5.1. Body weight and food consumption

During the 13 weeks of the treatment phase and 4 weeks of the recovery phase, the body weight of each rat was recorded, and the average food consumption was calculated weekly. At first, the amounts of feed were weighed before they were supplied to each cage, and their remnants were measured the next day. The differences were calculated and regarded as daily feed consumption per cage (g/cage/day). Then the weekly average food consumption per rat = the total consumption in a week per cage/5 (5 represented the number of rats per cage).

2.5.2. Hematological and coagulation analysis

At the end of the treatment phase or recovery phase, the blood from all rats in each group was collected after fasting for 16 h. The collected blood samples were assigned into three parts as follows: 1 mL blood with EDTA dipotassium salt (K2EDTA) anticoagulant, 1 mL blood with sodium citrate anticoagulant (anticoagulant: blood = 1:9), and 4–5 mL blood without anticoagulant.

The hematological analysis was performed to evaluate white blood cell (WBC) count, red blood cell (RBC) count, hemoglobin (HGB), hematocrit (HCT), mean corpuscular volume (MCV), mean corpuscular hemoglobin (MCH), MCH concentration (MCHC), platelet count (PLT), lymphocytes (LYM), monocytes (MON), neutrophils (NEU), eosinophils (EOS), basophils (BAS), and reticulocyte (RET) level. The prothrombin time (PT) and activated partial thromboplastin time (APTT) were also assessed.

2.5.3. Serum biochemistry analysis

For the serum biochemistry assay, dry tubes containing blood without anticoagulant were centrifuged at 3000 rpm for 15 min, and the supernatant was collected. The levels of alanine transaminase (ALT), aspartate transaminase (AST), total protein (TP), albumin (ALB), globulin (GLB), total bilirubin (TBIL), alkaline phosphatase (ALP), gamma-glutamyl transferase (GGT), glucose (GLU), undulin (UN), creatinine

(CRE), cholesterol (CHO), triglycerides (TG), creatine phosphokinase (CPK), ALB/GLB ratio (AGR), the potassium (K^+), sodium (Na^+), and chloride (Cl^-) ion concentrations were determined, and related parameters were also determined according to standard protocols (Zhao et al., 2020). The pH value was also determined for every sample.

2.5.4. Urine examination

At the end of the treatment phase or recovery phase, the urine samples were collected. Briefly, one rat was taken from each cage and given 5 mL of water by oral gavage, followed by close observation every 5–10 min. Once urinating, the urine was collected by a pipette into the test tube. The single acquisition time was no more than 1 h.

The following parameters, including the appearance, specific gravity (SG), pH value, leukocyte (LEU), nitrite (NIT), urinary protein (PRO), urinary GLU, ketonuria (KET), urobilinogen (UBG), urinary bilirubin (BIL), and erythrocyte (ERY), of the above-mentioned samples were examined and analyzed comparatively (Wang et al., 2019a).

2.5.5. Determination of organ weight and coefficients

At the end of the treatment phase or recovery phase, all the rats were sacrificed following the blood collection. The macroscopic appearance of the heart, liver, spleen, lung, kidney, brain, thymus, adrenal gland, uterus, ovary, testis, and epididymis were observed. Subsequently, all these organs were dissected, and the weights of the heart, liver, spleen, kidney, brain, thymus, adrenal gland, uterus, ovary, testis, and epididymis were recorded. The organ coefficients were calculated separately.

2.5.6. Histological analysis of organs

Immediately after weighing the reproductive organs obtained at the end of the treatment phase or recovery phase, their tissue fragments from the control group and high-dose group were fixed in 10% formaldehyde. These samples were dehydrated, embedded in paraffin, cut into 4- to 5- μ m-thick sections, and stained with hematoxylin and eosin (H&E) for microscopic histopathological examinations.

The pathological observations were performed using a microscope. The parameters of degeneration, necrosis, leukocyte infiltration, extravasation of blood, and congestion from different groups were

Table 1

Effect of three different doses of Qiguiyin on hematological and coagulation parameters in male or female rats at the end of the treatment phase or recovery phase.

Parameter (mean ± SD)	Treatment phase (n = 10)				Recovery phase (n = 5)			
	CN	TA	TB	TC	CN	TA	TB	TC
Male								
WBC ($\times 10^9/L$)	7.5 ± 1.3	8.6 ± 1.5	7.2 ± 1.3	7.7 ± 1.8	9.0 ± 2.8	7.0 ± 2.2	7.0 ± 2.0	5.8 ± 1.4
RBC ($\times 10^{12}/L$)	8.99 ± 0.32	8.76 ± 0.43	8.56 ± 0.32	8.14 ± 1.27	9.07 ± 0.20	8.71 ± 0.37*	8.49 ± 0.24**	8.80 ± 0.22
HGB (g/L)	158 ± 6	153 ± 5	146 ± 6**	149 ± 6**	153 ± 3	150 ± 9	148 ± 4	152 ± 6
HCT (L/L)	0.481 ± 0.018	0.467 ± 0.016	0.452 ± 0.016	0.436 ± 0.065**	0.460 ± 0.004	0.453 ± 0.029	0.452 ± 0.017	0.460 ± 0.017
MCV (fL)	53 ± 2	53 ± 2	53 ± 2	54 ± 3	51 ± 1	52 ± 1	53 ± 3	52 ± 1
MCH (pg)	17.6 ± 0.7	17.5 ± 0.5	17.1 ± 0.5	18.9 ± 4.6	16.9 ± 0.5	17.2 ± 0.6	17.4 ± 0.9	17.2 ± 0.3
MCHC (g/L)	328 ± 3	327 ± 3	324 ± 4	353 ± 84	332 ± 3	331 ± 5	326 ± 5	330 ± 4
PLT ($\times 10^9/L$)	898 ± 281	861 ± 116	858 ± 86	810 ± 130	799 ± 199	838 ± 254	768 ± 132	755 ± 131
LYM ($10^9/L$)	4.98 ± 0.79	5.76 ± 1.14	4.63 ± 1.27	4.64 ± 1.56	4.78 ± 1.68	4.17 ± 1.25	4.01 ± 1.12	3.46 ± 1.03
LYM (%)	66.8 ± 4.2	66.5 ± 5.7	63.9 ± 6.8	59.7 ± 11.2	53.2 ± 5.8	60.3 ± 8.1	57.4 ± 3.7	59.0 ± 3.7
MON ($10^9/L$)	0.02 ± 0.01	0.03 ± 0.01	0.02 ± 0.01	0.03 ± 0.02	0.05 ± 0.04	0.06 ± 0.03	0.05 ± 0.01	0.05 ± 0.03
MON (%)	0.2 ± 0.2	0.3 ± 0.2	0.2 ± 0.2	0.3 ± 0.2	0.6 ± 0.4	0.8 ± 0.3	0.7 ± 0.1	0.8 ± 0.3
NEU ($10^9/L$)	2.21 ± 0.61	2.55 ± 0.62	2.26 ± 0.37	2.74 ± 0.75	3.64 ± 1.14	2.30 ± 0.86*	2.60 ± 0.83	2.02 ± 0.31**
NEU (%)	29.3 ± 3.8	29.6 ± 5.2	32.4 ± 7.1	36.4 ± 9.6	40.7 ± 3.3	32.6 ± 4.1**	36.9 ± 3.2	35.5 ± 3.9*
EOS ($10^9/L$)	0.20 ± 0.08	0.23 ± 0.13	0.19 ± 0.07	0.20 ± 0.09	0.39 ± 0.25	0.35 ± 0.37	0.25 ± 0.08	0.18 ± 0.04
EOS (%)	2.8 ± 1.5	2.7 ± 1.6	2.6 ± 0.7	2.7 ± 1.5	4.4 ± 2.7	4.7 ± 4.5	3.6 ± 0.7	3.2 ± 0.3
BAS ($10^9/L$)	0.06 ± 0.02	0.07 ± 0.02	0.06 ± 0.02	0.07 ± 0.02	0.10 ± 0.04	0.11 ± 0.06	0.10 ± 0.03	0.09 ± 0.02
BAS (%)	0.8 ± 0.2	0.8 ± 0.2	0.8 ± 0.2	0.9 ± 0.3	1.1 ± 0.3	1.6 ± 0.8	1.4 ± 0.3	1.5 ± 0.2
RET ($10^{12}/L$)	0.222 ± 0.055	0.208 ± 0.058	0.225 ± 0.06	0.239 ± 0.086	0.320 ± 0.040	0.298 ± 0.038	0.292 ± 0.047	0.321 ± 0.047
RET (%)	1.23 ± 0.30	1.19 ± 0.33	1.32 ± 0.37	1.44 ± 0.40	1.76 ± 0.24	1.72 ± 0.24	1.72 ± 0.26	1.83 ± 0.3
PT (s)	17.5 ± 2.06	18.16 ± 3.60	18.89 ± 3.03	19.27 ± 6.04	16.94 ± 3.41	15.40 ± 2.34	14.94 ± 0.99	15.22 ± 1.17
APTT (s)	21.02 ± 1.08	21.75 ± 2.49	20.64 ± 0.93	21.38 ± 1.55	20.34 ± 0.38	20.96 ± 1.19	20.26 ± 0.18	20.82 ± 0.66
Female								
WBC ($\times 10^9/L$)	8.0 ± 2.2	7.2 ± 2.1	7.0 ± 1.7	7.7 ± 1.8	4.9 ± 1.9	4.2 ± 1.3	3.4 ± 1.1	4.2 ± 0.6
RBC ($\times 10^{12}/L$)	7.90 ± 0.36	8.03 ± 0.45	7.89 ± 0.33	7.77 ± 0.55	7.90 ± 0.33	7.89 ± 0.46	7.77 ± 0.35	7.94 ± 0.46
HGB (g/L)	146 ± 6	148 ± 7	146 ± 7	144 ± 8	143 ± 6	145 ± 6	142 ± 5	146 ± 5
HCT (L/L)	0.443 ± 0.017	0.449 ± 0.019	0.441 ± 0.020	0.438 ± 0.024	0.437 ± 0.011	0.446 ± 0.017	0.436 ± 0.018	0.447 ± 0.017
MCV (fL)	56 ± 2	56 ± 2	56 ± 2	56 ± 2	55 ± 1	56 ± 1	56 ± 2	56 ± 2
MCH (pg)	18.5 ± 0.6	18.4 ± 0.4	18.5 ± 0.5	18.6 ± 0.6	18.1 ± 0.4	18.5 ± 0.5	18.3 ± 0.8	18.4 ± 0.5
MCHC (g/L)	330 ± 3	330 ± 4	330 ± 6	330 ± 3	327 ± 4	326 ± 5	325 ± 3	327 ± 5
PLT ($\times 10^9/L$)	821 ± 159	674 ± 193	726 ± 100	735 ± 177	771 ± 178	721 ± 144	754 ± 66	783 ± 96
LYM ($10^9/L$)	5.82 ± 1.79	5.29 ± 1.55	5.21 ± 1.60	5.64 ± 1.61	2.97 ± 1.03	2.70 ± 0.81	2.25 ± 0.69	2.97 ± 0.28
LYM (%)	72.9 ± 5.7	73.6 ± 5.5	73.8 ± 8.9	72.1 ± 7.3	62.0 ± 7.7	63.5 ± 2.8	65.8 ± 5.5	71.3 ± 4.3
MON ($10^9/L$)	0.01 ± 0.01	0.01 ± 0.01	0.01 ± 0.01	0.01 ± 0.01	0.01 ± 0.01	0.01 ± 0.01	0.01 ± 0.01	0.00 ± 0.01
MON (%)	0.1 ± 0.1	0.1 ± 0.1	0.1 ± 0.1	0.1 ± 0.1	0.2 ± 0.1	0.2 ± 0.2	0.2 ± 0.2	0.1 ± 0.0
NEU ($10^9/L$)	1.80 ± 0.54	1.65 ± 0.77	1.55 ± 0.57	1.74 ± 0.49	1.71 ± 0.93	1.33 ± 0.36	1.06 ± 0.47	1.04 ± 0.29
NEU (%)	22.91 ± 4.42	22.86 ± 5.51	22.98 ± 8.38	22.96 ± 5.55	33.8 ± 6.4	31.8 ± 3.0	30.4 ± 5.4	24.4 ± 3.2**
EOS ($10^9/L$)	0.24 ± 0.23	0.16 ± 0.09	0.14 ± 0.06	0.27 ± 0.25	0.14 ± 0.09	0.16 ± 0.15	0.08 ± 0.03	0.12 ± 0.06
EOS (%)	2.9 ± 2.6	2.3 ± 1.3	2.0 ± 0.5	3.6 ± 3.6	2.9 ± 2.1	3.4 ± 2.0	2.3 ± 0.4	2.8 ± 1.0
BAS ($10^9/L$)	0.09 ± 0.03	0.08 ± 0.02	0.08 ± 0.02	0.09 ± 0.04	0.05 ± 0.01	0.04 ± 0.01	0.04 ± 0.02	0.06 ± 0.02
BAS (%)	1.2 ± 0.4	1.2 ± 0.5	1.1 ± 0.5	1.2 ± 0.6	1.1 ± 0.4	1.0 ± 0.4	1.3 ± 0.3	1.4 ± 0.5
RET ($10^{12}/L$)	0.229 ± 0.026	0.245 ± 0.042	0.231 ± 0.042	0.267 ± 0.070	0.289 ± 0.030	0.258 ± 0.027	0.269 ± 0.040	0.304 ± 0.043
RET (%)	1.45 ± 0.15	1.53 ± 0.24	1.47 ± 0.29	1.72 ± 0.40	1.83 ± 0.19	1.64 ± 0.22	1.73 ± 0.22	1.92 ± 0.32
PT (s)	12.75 ± 1.02	13.19 ± 0.62	12.59 ± 0.55	13.02 ± 0.61	12.46 ± 0.56	13.48 ± 2.99	12.44 ± 0.88	13.16 ± 0.84
APTT (s)	21.50 ± 2.93	21.88 ± 1.99	20.88 ± 1.18	20.51 ± 0.91	21.02 ± 0.56	20.48 ± 0.33	20.22 ± 0.22	20.26 ± 0.17

Significantly different from CN group: * $p < 0.05$, ** $p < 0.01$.

compared and recorded.

2.6. Statistical analysis

Data were statistically compared using statistical analysis system (SAS) for Windows (version 9.4; Order Number: 9CFYF4). Various parameters were statistically evaluated by the Student's t-test and expressed as the mean ± standard deviation. Significant differences were classified as: *for $p < 0.05$; or more significant **for $p < 0.01$.

3. Results

3.1. General toxicity evaluation

No significant mortality and noticeable clinical signs of treatment-related toxicity were recorded throughout the 17 weeks. The data suggested that the Qiguiyin treatment did not induce any significant morphological changes.

3.1.1. Body weight and food consumption

The changes in body weight are an important indicator of the adverse effects of plant drugs. Before the treatment, the body weight was 185.6–226.8 g in males and 186.0–218.7 g in females. Fig. 2A and B shows that during the 17 weeks of the experimental period, the body weight was gradually and significantly increased from the 1st week to the 17th week in all these groups compared with their corresponding baseline values, regardless of the gender. There was no significant difference in the body weight between the experimental groups and control group.

Fig. 2C and D reveal that there was a consistent variation trend of the average food consumption and body weight among different groups. Moreover, the variation trend was similar in both sexes.

3.1.2. Hematological and coagulation results

The hematological parameters are one of the most sensitive targets for toxicity, serving as an important index of health status for both humans and animals. Table 1 lists the effect of Qiguiyin on the hematological parameters of the experimental groups and control group.

The results indicated that all hematological parameters remained

Table 2

Effect of three different doses of Qiguiyin on serum biochemical parameters in male or female rats at the end of the treatment phase or recovery phase. Color images are available online.

Parameter (mean $\pm$ SD)	Treatment phase (n = 10)				Recovery phase (n = 5)			
	CN	TA	TB	TC	CN	TA	TB	TC
Male								
ALT (U/L)	58 $\pm$ 39	45 $\pm$ 17	43 $\pm$ 10		49 $\pm$ 14	52 $\pm$ 16	48 $\pm$ 8	56 $\pm$ 31
AST (U/L)	145 $\pm$ 61	137 $\pm$ 36	128 $\pm$ 42		135 $\pm$ 25	165 $\pm$ 21	164 $\pm$ 17	165 $\pm$ 47
TP (g/L)	63.47 $\pm$ 3.00	61.93 $\pm$ 3.71	60.65 $\pm$ 2.35		61.21 $\pm$ 3.21	59.62 $\pm$ 3.43	59.99 $\pm$ 1.96	56.68 $\pm$ 2.44
ALB (g/L)	29.57 $\pm$ 1.49	28.88 $\pm$ 1.76	29.18 $\pm$ 1.47		29.46 $\pm$ 1.48	27.67 $\pm$ 1.09	27.11 $\pm$ 0.92	27.28 $\pm$ 0.78
GLB (g/L)	33.90 $\pm$ 2.01	33.05 $\pm$ 2.43	31.47 $\pm$ 1.27*		31.75 $\pm$ 2.10*	31.96 $\pm$ 3.23	32.89 $\pm$ 2.18	29.40 $\pm$ 1.84
TBIL (μ mol/L)	2.52 $\pm$ 0.31	2.24 $\pm$ 0.38	2.34 $\pm$ 0.34		2.23 $\pm$ 0.11	2.27 $\pm$ 0.30	2.17 $\pm$ 0.30	2.45 $\pm$ 0.35
ALP (U/L)	73 $\pm$ 17	73 $\pm$ 10	72 $\pm$ 13		81 $\pm$ 19	85 $\pm$ 19	80 $\pm$ 15	77 $\pm$ 18
GGT (U/L)	0.4 $\pm$ 0.7	0.5 $\pm$ 0.7	0.6 $\pm$ 0.8		0.6 $\pm$ 0.7	0.8 $\pm$ 0.4	0.4 $\pm$ 0.5	0.0 $\pm$ 0.0
GLU (mmol/L)	7.91 $\pm$ 0.94	6.68 $\pm$ 0.84	7.51 $\pm$ 1.18		7.31 $\pm$ 1.65	8.30 $\pm$ 1.16	8.33 $\pm$ 2.09	7.82 $\pm$ 0.57
UN (mmol/L)	6.37 $\pm$ 0.31	6.40 $\pm$ 0.54	6.11 $\pm$ 0.66		6.19 $\pm$ 0.73	6.33 $\pm$ 1.21	5.26 $\pm$ 0.21	5.92 $\pm$ 0.66
CRE (μ mol/L)	26.78 $\pm$ 1.33	26.38 $\pm$ 4.10	25.54 $\pm$ 2.53		25.07 $\pm$ 3.52	31.61 $\pm$ 4.26	27.38 $\pm$ 3.26	27.36 $\pm$ 3.45
CHO (mmol/L)	2.13 $\pm$ 0.42	1.61 $\pm$ 0.29**	1.97 $\pm$ 0.35		1.81 $\pm$ 0.26*	2.09 $\pm$ 0.43	1.84 $\pm$ 0.46	1.89 $\pm$ 0.19
TG (mmol/L)	0.50 $\pm$ 0.19	0.35 $\pm$ 0.14	0.47 $\pm$ 0.20		0.56 $\pm$ 0.38	0.31 $\pm$ 0.11	0.50 $\pm$ 0.33	0.36 $\pm$ 0.16
CPK (U/L)	662 $\pm$ 258	765 $\pm$ 279	634 $\pm$ 334		622 $\pm$ 175	945 $\pm$ 287	998 $\pm$ 108	971 $\pm$ 277
AGR	0.87 $\pm$ 0.05	0.88 $\pm$ 0.06	0.93 $\pm$ 0.05*		0.93 $\pm$ 0.05*	0.87 $\pm$ 0.09	0.83 $\pm$ 0.08	0.93 $\pm$ 0.04
K (mmol/L)	4.76 $\pm$ 0.34	4.80 $\pm$ 0.36	4.91 $\pm$ 0.41		4.95 $\pm$ 0.57	4.86 $\pm$ 0.23	4.82 $\pm$ 0.48	4.80 $\pm$ 0.46
Na ⁺ (mmol/L)	142.10 $\pm$ 0.88	142.10 $\pm$ 1.20	141.60 $\pm$ 1.07		141.30 $\pm$ 1.42	140.20 $\pm$ 0.84	141.00 $\pm$ 1.22	140.00 $\pm$ 1.87
Cl ⁻ (mmol/L)	106.70 $\pm$ 1.42	106.80 $\pm$ 1.69	107.20 $\pm$ 1.23		106.80 $\pm$ 1.32	107.00 $\pm$ 1.22	107.20 $\pm$ 0.84	107.20 $\pm$ 1.48
Female								
ALT (U/L)	43 $\pm$ 20	42 $\pm$ 14	43 $\pm$ 33	40 $\pm$ 11	68 $\pm$ 48	75 $\pm$ 52	93 $\pm$ 129	49 $\pm$ 16
AST (U/L)	135 $\pm$ 94	118 $\pm$ 30	113 $\pm$ 67	113 $\pm$ 41	158 $\pm$ 125	175 $\pm$ 95	173 $\pm$ 182	116 $\pm$ 36
TP (g/L)	69.77 $\pm$ 6.17	67.84 $\pm$ 5.97	68.24 $\pm$ 5.49	67.14 $\pm$ 3.14	64.11 $\pm$ 3.98	66.65 $\pm$ 5.93	66.70 $\pm$ 6.04	61.56 $\pm$ 4.84
ALB (g/L)	36.96 $\pm$ 4.63	36.57 $\pm$ 3.86	36.50 $\pm$ 3.87	35.93 $\pm$ 2.64	33.48 $\pm$ 2.37	37.18 $\pm$ 3.78	37.41 $\pm$ 3.93	33.09 $\pm$ 3.25
GLB (g/L)	32.81 $\pm$ 1.96	31.28 $\pm$ 2.37	31.74 $\pm$ 2.43	31.20 $\pm$ 2.02	30.63 $\pm$ 1.90	29.47 $\pm$ 2.26	29.28 $\pm$ 3.09	28.47 $\pm$ 1.75
TBIL (μ mol/L)	3.58 $\pm$ 0.60	3.42 $\pm$ 0.59	3.26 $\pm$ 0.65	3.19 $\pm$ 0.31	2.49 $\pm$ 0.43	3.18 $\pm$ 0.46*	2.94 $\pm$ 0.41	2.37 $\pm$ 0.11
ALP (U/L)	32 $\pm$ 8	37 $\pm$ 9	30 $\pm$ 7	32 $\pm$ 6	30 $\pm$ 4	36 $\pm$ 9	25 $\pm$ 8	31 $\pm$ 8
GGT (U/L)	0.4 $\pm$ 0.5	0.7 $\pm$ 0.7	0.7 $\pm$ 0.7	1.0 $\pm$ 0.7	1.0 $\pm$ 0.0	1.2 $\pm$ 0.8	0.6 $\pm$ 0.5	0.6 $\pm$ 0.5
GLU (mmol/L)	6.95 $\pm$ 1.16	7.33 $\pm$ 1.15	7.49 $\pm$ 0.93	6.61 $\pm$ 0.74	8.28 $\pm$ 1.50	8.17 $\pm$ 0.97	8.40 $\pm$ 0.50	7.43 $\pm$ 1.60
UN (mmol/L)	5.91 $\pm$ 0.78	6.09 $\pm$ 0.65	5.90 $\pm$ 0.61	6.33 $\pm$ 0.68	5.12 $\pm$ 0.70	5.84 $\pm$ 0.89	5.16 $\pm$ 0.86	5.63 $\pm$ 1.21
CRE (μ mol/L)	32.37 $\pm$ 3.70	32.34 $\pm$ 2.90	29.89 $\pm$ 3.95	30.76 $\pm$ 3.54	30.46 $\pm$ 2.43	30.76 $\pm$ 1.93	29.50 $\pm$ 5.20	30.12 $\pm$ 2.69
CHO (mmol/L)	2.17 $\pm$ 0.66	2.10 $\pm$ 0.35	2.32 $\pm$ 0.46	2.04 $\pm$ 0.28	2.37 $\pm$ 0.37	2.28 $\pm$ 0.57	2.48 $\pm$ 0.43	1.97 $\pm$ 0.62
TG (mmol/L)	0.41 $\pm$ 0.20	0.39 $\pm$ 0.17	0.51 $\pm$ 0.26	0.32 $\pm$ 0.11	0.49 $\pm$ 0.06	0.40 $\pm$ 0.18	0.47 $\pm$ 0.16	0.23 $\pm$ 0.07**
CPK (U/L)	324 $\pm$ 112	411 $\pm$ 196	341 $\pm$ 109	393 $\pm$ 102	366 $\pm$ 175	374 $\pm$ 147	273 $\pm$ 172	459 $\pm$ 267
AGR	1.13 $\pm$ 0.10	1.17 $\pm$ 0.07	1.15 $\pm$ 0.11	1.16 $\pm$ 0.12	1.09 $\pm$ 0.05	1.26 $\pm$ 0.05**	1.28 $\pm$ 0.13	1.16 $\pm$ 0.06
K ⁺ (mmol/L)	4.10 $\pm$ 0.55	4.61 $\pm$ 1.14	4.29 $\pm$ 0.31	4.18 $\pm$ 0.30	3.84 $\pm$ 0.29	4.26 $\pm$ 0.66	3.68 $\pm$ 0.41	4.20 $\pm$ 0.36
Na ⁺ (mmol/L)	142.80 $\pm$ 1.48	142.00 $\pm$ 1.15	141.2 $\pm$ 0.92**	141.1 $\pm$ 1.52**	141.20 $\pm$ 0.84	141.00 $\pm$ 1.87	142.00 $\pm$ 2.35	140.80 $\pm$ 1.10
Cl ⁻ (mmol/L)	105.90 $\pm$ 1.97	107.60 $\pm$ 1.90	108.10 $\pm$ 3.41	106.60 $\pm$ 1.17	107.80 $\pm$ 1.64	107.80 $\pm$ 1.48	107.60 $\pm$ 2.19	107.80 $\pm$ 0.84

Significantly different from CN group: * p < 0.05, ** p < 0.01.

within the physiological range throughout the 17 weeks of the experimental period. There were no obvious changes in WBC, RBC, HGB, HCT, MCV, MCH, MCHC, PLT, LYM, MON, NEU, EOS, BAS, PT, and APTT among the control group and three experimental groups. Moreover, no dose-dependent relationship was found in the three experimental groups. Besides, there were no significant differences in these parameters between male and female rats.

The results indicated that all hematological parameters remained within the physiological range throughout the 17 weeks of the experimental period. At the end of the treatment phase, the HGB of male rats in the TB or TC group was lower compared with the control group (p < 0.01, p < 0.01), and the HCT level of male rats in the TC group was lower compared with the control group (p < 0.01). However, at the end of the recovery phase, the RBC count of male rats in the TA or TB group was lower compared with the control group (p < 0.05, p < 0.01), the NEU ($10^9/L$) of male rats in the TA or TC group was lower compared with the control group (p < 0.05, p < 0.01), the NEU(%) of male rats in the TA or TC group was also lower compared with the control group (p < 0.01, p < 0.05). It was noting that all these above-mentioned differences were statistically significant but not meaningful in clinical practice, because all these parameters were in normal physiological range. There were no obvious changes in WBC, HCT, MCV, MCH, MCHC, PLT, LYM, MON, EOS, BAS, PT, and APTT among the control group and three experimental groups. Moreover, no dose-dependent relationship was found in the three experimental groups. Besides, there were no significant differences in these parameters between male and female rats.

3.1.3. Serum biochemical results

Another important aspect analyzed in the toxicological evaluation was the serum biochemical parameters. Table 2 summarizes the data of the biochemical parameters in rats at the end of the treatment phase or recovery phase.

All the serum biochemical parameters of each group remained within the physiological range throughout the 17 weeks of the experimental period. At the end of the treatment phase, GLB, CHO, AGR, and Na⁺ were significantly different in male rats between the TA group and control group (p < 0.01). At the end of the recovery phase, the TBIL and AGR in the TA group were higher compared with the control group (p < 0.01), while the TG in the TC group was lower compared with the control group in female rats (p < 0.01). However, these differences were not clinically significant. Especially, there were no differences in ALB, ALP, GLU, UN, CRE, K⁺, Na⁺, Cl⁻ ion concentrations, ALT, AST, and TP between the two groups. Furthermore, there was no gender-related difference in these parameters.

3.1.4. Urinary indices

Urinary indices can reflect the metabolic status of the body, serving as the wind vane of liver and kidney function. Table 3 shows that all the urinary parameters from each group were within the normal physiological range throughout the 17 weeks of the experimental period. Especially, there were no significant differences in SG, pH value, and GLU among different groups or between different genders.

Table 3

Effect of three different doses of Qiguiyin on urinary indices in male or female rats at the end of the treatment phase or recovery phase. Color images are available online.

Index (mean $\pm$ SD)	Treatment phase (n = 10)				Recovery phase (n = 5)			
	CN	TA	TB	TC	CN	TA	TB	TC
Male								
Appearance	Light yellow	Light yellow	Light yellow (7) Dark yellow (3)	Light yellow	Light yellow	Light yellow	Light yellow	Light yellow
SG	1.014 $\pm$ 0.003	1.012 $\pm$ 0.003	1.013 $\pm$ 0.003	1.014 $\pm$ 0.002	1.011 $\pm$ 0.002	1.011 $\pm$ 0.002	1.011 $\pm$ 0.002	1.011 $\pm$ 0.002
pH	8.20 $\pm$ 0.42	8.20 $\pm$ 0.42	8.00 $\pm$ 0.43	8.25 $\pm$ 0.46	7.80 $\pm$ 0.45	8.20 $\pm$ 0.45	7.80 $\pm$ 0.45	8.20 $\pm$ 0.45
LEU	-	-	-	-	-	-	-	-
NIT	-	-	-	-	-	-	-	-
PRO	-	-	-	-	-	-	-	-
GLU	-	-	-	-	-	-	-	-
KET	-	-	-	-	-	-	-	-
UBG	-	-	-	-	-	-	-	-
BIL	-	-	-	-	-	-	-	-
ERY	-	-	-	-	-	-	-	-
Female								
Appearance	Light yellow	Light yellow	Light yellow	Light yellow	Light yellow	Light yellow	Light yellow	Light yellow
SG	1.014 $\pm$ 0.004	1.010 $\pm$ 0.003	1.011 $\pm$ 0.004	1.009 $\pm$ 0.004	1.014 $\pm$ 0.004	1.012 $\pm$ 0.003	1.012 $\pm$ 0.003	1.011 $\pm$ 0.002
pH	7.80 $\pm$ 0.60	7.60 $\pm$ 0.80	7.60 $\pm$ 1.00	7.10 $\pm$ 0.40	7.60 $\pm$ 0.89	8.20 $\pm$ 0.45	8.00 $\pm$ 0.00	7.80 $\pm$ 0.45
LEU	-	-	-	-	-	-	-	-
NIT	-	-	-	-	-	-	-	-
PRO	-	-	-	-	-	-	-	-
GLU	-	-	-	-	-	-	-	-
KET	-	-	-	-	-	-	-	-
UBG	-	-	-	-	-	-	-	-
BIL	-	-	-	-	-	-	-	-
ERY	-	-	-	-	-	-	-	-

1No significantly different from CN group. 2. “-” negative, “+” positive.3. The number in brackets is the rat count.

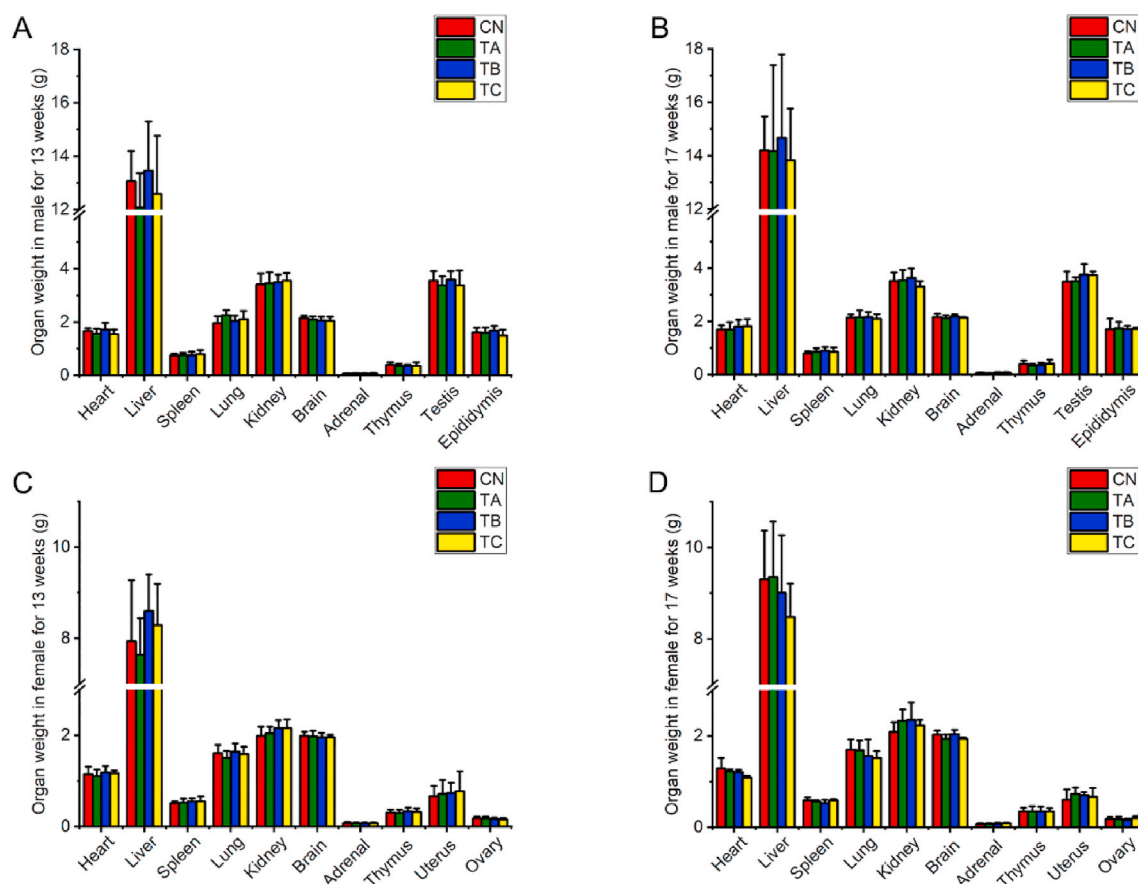


Fig. 3. Effects of three different doses of Qiguiyin on absolute organ weights in male or female rats at the end of the treatment phase or recovery phase. In the treatment phase, n = 20 in each group, including 10 males (A) and 10 females (B). In the recovery phase, n = 10 in each group, including five males (C) and five females (D). Compare with the control group. *P < 0.05, **P < 0.01. Color images are available online. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

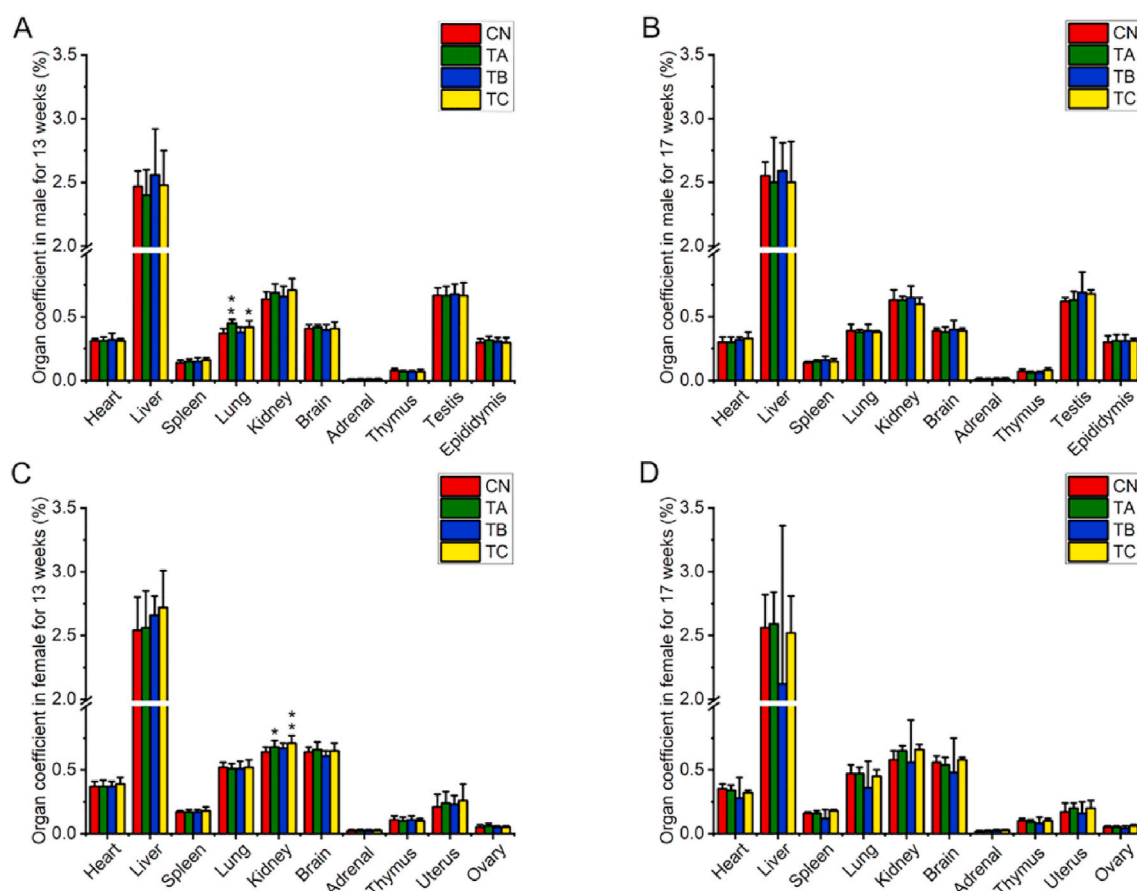


Fig. 4. Effects of three different doses of Qiguiyin on relative organ coefficients in male or female rats at the end of the treatment phase or recovery phase. In the treatment phase, $n = 20$ in each group, including 10 males (A) and 10 females (B). In the recovery phase, $n = 10$ in each group, including five males (C) and five females (D). Compare with the control group. * $P < 0.05$, ** $P < 0.01$. Color images are available online. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

3.1.5. Organ absolute weight and coefficients

Organ-to-body weight ratio (organ coefficients) is another sensitive index for toxicity evaluation. At the end of the treatment phase or recovery phase, the analyses of the absolute organ weights (Fig. 3) and related organ coefficients (Fig. 4) revealed no significant differences in the heart, spleen, and liver of rats compared with the control group, regardless of the gender. The male lung coefficient of the TA or TC group was higher compared with the control group ($P < 0.01$, $P < 0.05$, respectively), and the female renal coefficient of the TA or TC group was higher compared with the control group ($P < 0.05$, $P < 0.01$, respectively). However, we believed that these differences did not reflect biological significances, because the appearance and absolute weight of the lung and kidney in the TA and TC groups were normal, and the histopathological results were also normal. Moreover, the lung and kidney coefficients in the TB group were normal, and there was no dose-dependent relationship.

3.1.6. Histopathological results

Fig. 5A shows that there was one case of alveolar macrophage aggregation (1/20), one case of focal kidney tubular degeneration and regeneration (1/20), and one case of prostatic stromal inflammatory cell infiltration (1/10) in the control group. There were two cases of focal myocardial necrosis and inflammatory cell infiltration (2/20), and one case of interstitial focal liver inflammatory cell infiltration (1/20) in the TA group. There was one case of heart focal inflammatory cell infiltration (1/20), one case of pulmonary hemorrhage (1/20), and one case of prostatic stromal inflammatory cell infiltration (1/10) in the TB group. There was one case of focal kidney tubular degeneration and

regeneration (1/20) in the TC group.

There was one case of alveolar macrophage aggregation (1/20), and one case of focal hemorrhage of hepatocytes in the control group. There was one case of focal cardiomyocyte degeneration and necrosis with inflammatory cell infiltration (1/20), one case of ovarian atrophy (1/10), and one case of prostatic stromal inflammatory cell infiltration (1/10) in the TA group. There was one case of focal glomerular and tubular atrophy (1/20) in the TB group. There was one case of interstitial pneumonia (1/20) in the TC group (Fig. 5B).

Some atrophied seminiferous tubules and decreased spermatogenic cells were found in two cases (2/10) in the TC group only at the end of the treatment phase, while nothing was found in other groups no matter at the end of the treatment phase or recovery phase. Particularly, at the end of the recovery phase, the detectable abnormalities of the testis in the treatment phase were not detected again in the TC group (Fig. 6). Moreover, no obvious changes were found in the reproductive organs of the female rats throughout the experiment. Besides, no obvious pathological changes were found in other organs.

In short, we found that no matter in the control group or the experimental groups, all the organ coefficients were within the physiological range, and there were no statistical differences between the control group and experimental groups. Especially, there were no obvious pathological changes in all groups. Therefore, the above-mentioned incidental pathological changes in some organs were not attributed to toxicological reactions, no matter in the control group or TC group. Only two cases of male testicular seminiferous tubule lesions were considered as transient pathological changes related to toxicity.

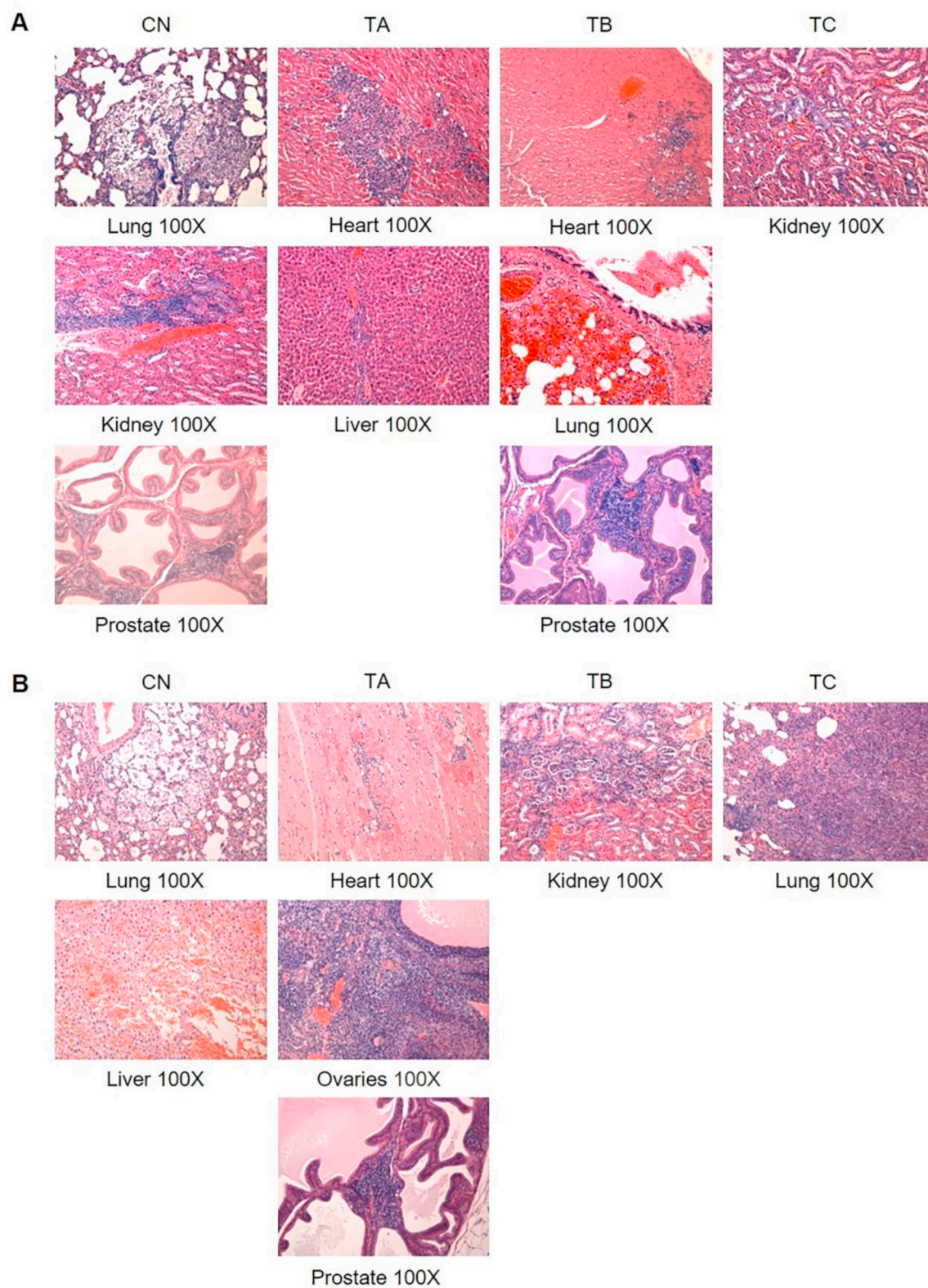


Fig. 5. Histopathological results of organs in each group. (A) Results at the end of the treatment phase; (B) Results at the end of the recovery phase. All photos under 100 folds.

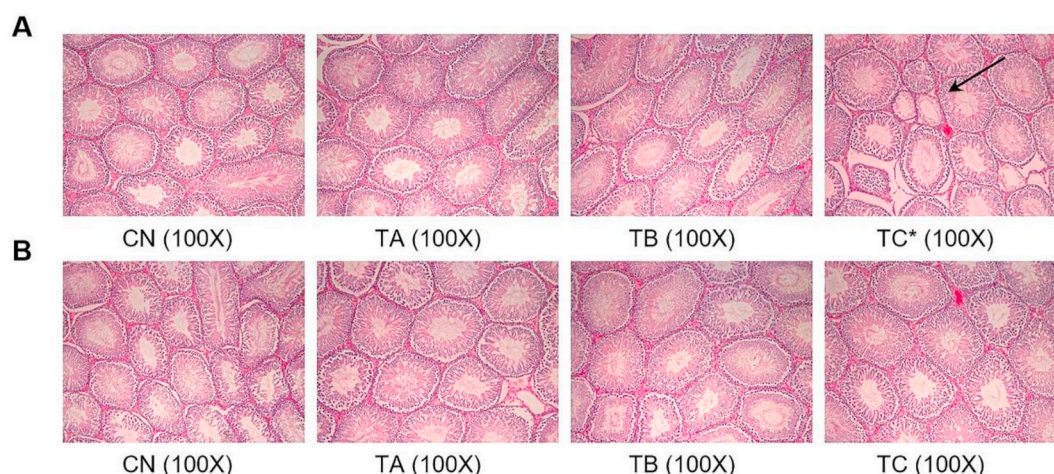


Fig. 6. Histopathological results of testis in each group. (A) Results at the end of the treatment phase; □ atrophied seminiferous tubules, *positive histopathological result (B) Results at the end of the recovery phase. All photos under 100 folds.

4. Discussion

Herbal medicines have been used in both traditional and non-traditional forms of medicine for thousands of years and are considered safe by the general population. However, their potential toxicity to patients has become a serious medical issue (Li et al., 2018b; Parvez and Rishi, 2019). Qiguiyin, a traditional Chinese medicinal formula, has been recorded as a hospital preparation. Elucidation of its possible toxicity to each system may provide experimental basis for the secondary development of Qiguiyin.

Based on the immunomodulatory effect of Qiguiyin in our previous study, Qiguiyin can delay the drug resistance of *P. aeruginosa* (Ding et al., 2020). To further clarify the core active pharmacological components of Qiguiyin, different extraction methods were used in this study. It has been reported that both water extraction and alcohol extraction are effective methods for drug separation (Guo et al., 2019; Miao et al., 2019; Yuen et al., 2021). Our study confirmed that there were more active components from the water extraction compared with the alcohol extraction, which was consistent with Cai's study (Cai et al., 2016). Moreover, different parameters of HPLC methods were also optimized. These explorations helped us construct a standard process of Qiguiyin extraction preparation.

Qiguiyin, a hospital preparation, is composed of *A. hamosus*, *A. sinensis*, *L. sempervirens*, *A. annua*, and *P. cuspidatum*. At present, studies have confirmed that these components are safe and non-toxic separately. Song has found that in the toxicity study of HT042 (*A. hamosus* extract mixture) and each of the herbs, no deaths occur, and there is no indication of toxicity from clinical signs, body weight, and necropsy findings (Song et al., 2017). Another study has shown that there are no significant adverse effects of the AG (*A. sinensis* extract) on food consumption, body weight, mortality, hematology, biochemistry, necropsy, organ weight, and histopathology throughout the study in male and female rats (Yun et al., 2015). Moreover, it has been found that GCWB104 (*L. sempervirens* extract) safely improves irritable bowel syndrome symptoms (Choi et al., 2020), and SPB-201 (*A. annua* extract) has efficacy and safety in liver function improvement (Han et al., 2020).

In the present study, rats were consecutively treated with 10, 30, and 50 times of human doses of Qiguiyin for 13 weeks (about 3 months). We found no significant pathological changes in any organs or tissues even when 50 times of human doses of Qiguiyin were used. During the experiment, the pathological changes in some organs, such as the heart, liver, lung, and kidney, were not related to toxicology. Only some atrophied seminiferous tubules and decreased spermatogenic cells were transiently found in two (2/10) male rats, while they were cleared away at the end of the recovery phase. These findings were consistent with the

latest research (Wang et al., 2019b). Wang has found that prostate and seminal vesicles may also be the target organs of fructus psoraleae (FP)-induced toxicity. Moreover, abnormal hormone and enzyme production related to the hypothalamic-pituitary-adrenal (HPA) axis from ethanol (EtOH) extract of FP may be the potential toxic mechanism for changes in the adrenal gland and sex organs of male rats. In the toxicological study of arecae semen aqueous extract (ASAE), the relative organ weights of testis in male rats in the 4500 mg/kg group are significantly different from the control group at the end of the treatment period (Lin et al., 2018). Therefore, TCM may have temporary or transient effects on the reproductive system. To prevent irreversible reproductive effects, TCM should not be taken for a long time.

In addition, transient changes were seen in the hematological and biochemical parameters, such as HGB of male rats from TB and TC groups during the treatment phase, RBC of male rats from TA and TB groups, and NEU of male rats from TA and TC groups during the recovery phase, and so on. Similar changes were found in GLB, AGR of male rats from TB and TC groups during the treatment phase, CHO of male rats from TA and TC groups during the treatment phase, and Na⁺ of female rats from TB and TC groups during the treatment phase. These alterations were not related to Qiguiyin treatment because they remained within normal ranges throughout the 17 weeks, no matter in the treatment phase or recovery phase.

Taken together, although Qiguiyin can be used in clinical practice, no comprehensive safety evaluation has been performed regarding its toxicological profile in animals. We, for the first time, evaluated the toxicological safety of Qiguiyin. Therefore, our findings provided valuable insights into the new clinical practice of the Qiguiyin formula.

5. Conclusions

In this 17-week repeated-dose subchronic toxicity experiment, we showed no significant pathological changes in any organs or tissues at three different doses, regardless of the gender, except for transient testicular seminiferous tubule lesions in male rats. Collectively, Qiguiyin was safe and had no distinct toxicity or side effects. All these results laid an experimental foundation for the secondary development of Qiguiyin in clinical application.

Author contributions

Junying Ding performed experiments, conducted data analysis, and drew the manuscript. Fan Zhang, Yunxin Zhou, and Siyang Li contributed to the tests. Xiang Gao and Youran Lu took part in the animal experiments. Qiguiyin came from Qingquan Liu with TCM theoretical

system.

Declaration of competing interest

The authors declare that there are no conflicts of interest in this study.

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References

- Bahaeddin, Z., Yans, A., Khodaghali, F., Sahranavard, S., 2018. Dietary supplementation with Allium hirtifolium and/or Astragalus hamosus improved memory and reduced neuro-inflammation in the rat model of Alzheimer's disease. *Appl. Physiol. Nutr. Metabol.* 43, 558–564. <https://doi.org/10.1139/apnm-2017-0585>.
- Cai, H.D., Su, S.L., Li, Y.H., Zhu, Z.H., Guo, J.M., Zhu, Y., Guo, S., Qian, D.W., Duan, J.N., McPhee, J., 2016. Simultaneous determination of four tanshinones by UPLC-TQ/MS and their pharmacokinetic application after administration of single ethanol extract of danshen combined with water extract in normal and adenine-induced chronic renal failure rats. *Molecules* 21, 1630. <https://doi.org/10.3390/molecules21121630>.
- Cheng, C.Y., Kao, S.T., Lee, Y.C., 2020. Angelica sinensis extract protects against ischemia-reperfusion injury in the hippocampus by activating p38 MAPK-mediated p90RSK/p-Bad and p90RSK/CREB/BDNF signaling after transient global cerebral ischemia in rats. *J. Ethnopharmacol.* 252, 112612. <https://doi.org/10.1016/j.jep.2020.112612>.
- Choi, Y., Kim, N., Noh, G.T., Lee, J.Y., Lee, D.H., 2020. The efficacy and safety of GCWB104 (Flos Lonicera extract) in functional dyspepsia: a single-center, randomized, double-blind, placebo-controlled study. *Gut Liver* 14, 67–78. <https://doi.org/10.5009/gnl19283>.
- Choy, Y.M., Leung, K.N., Cho, C.S., Wong, C.K., Pang, P.K., 1994. Immunopharmacological studies of low molecular weight polysaccharide from Angelica sinensis. *Am. J. Chin. Med.* 22, 137–145. <https://doi.org/10.1142/s0192415x94000176>.
- Ding, J.Y., Ding, X.F., Lu, Y.R., An, S.D., Cui, X.R., Liu, Q.Q., 2019. Preliminary study on the immunoregulatory effect of Qiguiyin on Pseudomonas aeruginosa-induced pneumonia in rats. *J. Hebei TCM Pharmacol.* 34, 1–5. <https://doi.org/10.16370/j.cnki.13-1214/r.2019.02.001>.
- Ding, J.Y., Gao, X., Gui, H., Ding, X.F., Lu, Y.R., An, S.D., Liu, Q.Q., 2020. Proteomic analysis of proteins associated with inhibition of Pseudomonas aeruginosa resistance to imipenem mediated by the Chinese herbal medicine Qi Gui Yin. *Microb. Drug Resist.* <https://doi.org/10.1089/mdr.2020.0110>.
- Gao, H., Dou, L.R., Shan, L., Sun, Y., Li, W., 2018a. Proliferation and committed differentiation into dopamine neurons of neural stem cells induced by the active ingredients of radix astragal. *Neuroreport* 29, 577–582. <https://doi.org/10.1097/wnr.0000000000000997>.
- Gao, S.M., Liu, J.S., Wang, M., Cao, T.T., Qi, Y.D., Zhang, B.G., Sun, X.B., Liu, H.T., Xiao, P.G., 2018b. Traditional uses, phytochemistry, pharmacology and toxicology of Codonopsis: a review. *J. Ethnopharmacol.* 219, 50–70. <https://doi.org/10.1016/j.jep.2018.02.039>.
- Guo, N., Ping, K., Jiang, Y.W., Wang, L.T., Niu, L.J., Liu, Z.M., Fu, Y.J., 2019. Natural deep eutectic solvents couple with integrative extraction technique as an effective approach for mulberry anthocyanin extraction. *Food Chem.* 296, 78–85. <https://doi.org/10.1016/j.foodchem.2019.05.196>.
- Han, B., Kim, S.M., Nam, G.E., Kim, S.H., Park, S.J., Park, Y.K., Baik, H.W., 2020. A randomized, double-blind, placebo-controlled, multi-centered clinical study to evaluate the efficacy and safety of Artemisia annua extract for improvement of liver function. *Clin Nutr Res* 9, 258–270. <https://doi.org/10.7762/cnr.2020.9.4.258>.
- Kim, Y.J., Lee, J.Y., Kim, H.J., Kim, D.H., Lee, T.H., Kang, M.S., Park, W., 2018. Anti-inflammatory effects of angelica sinensis (Oliv.) Diels water extract on RAW 264.7 induced with lipopolysaccharide. *Nutrients* 10, 647. <https://doi.org/10.3390/nu10050647>.
- Kong, L.B., Ma, Q., Gao, J., Qiu, G.S., Wang, L.X., Zhao, S.M., Bao, Y.G., Liu, Q.Q., 2015. Effect of Qiguiyin Decoction on multidrug-resistant Pseudomonas aeruginosa infection in rats. *Chin. J. Integr. Med.* 21, 916–921. <https://doi.org/10.1007/s11655-015-2089-2>.
- Li, J.X., Xu, L., Sang, R., Yu, Y.F., Ge, B.J., Zhang, X.M., 2018a. Immunomodulatory and anti-inflammatory effects of total flavonoids of Astragalus by regulating NF-KB and MAPK signalling pathways in RAW 264.7 macrophages. *Pharmazie* 73, 589–593. <https://doi.org/10.1691/ph.2018.8633>.
- Li, X., Zhou, J., Dou, Y., Shi, Y., Wang, Y., Hong, J., Zhao, J., Zhang, J., Yuan, Y., Zhou, M., Wei, X., 2020. The protective effects of angelica organic acid against ox-LDL-induced autophagy dysfunction of HUVECs. *BMC Complement Med Ther* 20, 164. <https://doi.org/10.1186/s12906-020-02968-7>.
- Li, X.J., Li, X.Y., Huang, N.N., Liu, R.P., Sun, R., 2018b. A comprehensive review and perspectives on pharmacology and toxicology of saikosaponins. *Phytomedicine* 50, 73–87. <https://doi.org/10.1016/j.phymed.2018.09.174>.
- Li, Z.X., Zhao, G.D., Xiong, W., Linghu, K.G., Ma, Q.S., Cheang, W.S., Yu, H., Wang, Y., 2019. Immunomodulatory effects of a new whole ingredients extract from Astragalus: a combined evaluation on chemistry and pharmacology. *Chin. Med.* 14, 12. <https://doi.org/10.1186/s13020-019-0234-0>.
- Lin, B., Cai, B., Wang, H., 2019. Honeysuckle extract relieves ovalbumin-induced allergic rhinitis by inhibiting AR-induced inflammation and autoimmunity. *Biosci. Rep.* 39, BSR20190673. <https://doi.org/10.1042/bsr20190673>.
- Lin, Q.H., Jia, Z., Xu, X.F., Xu, S.Y., Han, T., Gao, Y., Zhang, Y., Zhang, H., Liu, H., Li, J., Li, X.R., 2018. Sub-chronic toxicity study of arecae semen aqueous extract in Wistar rats. *J. Ethnopharmacol.* 215, 176–183. <https://doi.org/10.1016/j.jep.2017.08.031>.
- Ma, T.T., Feng, X.Z., Wang, X.Y., 2017. Effects and mechanism of Angelica Sinensis Radix on Th1/Th2 and Th17/Treg in mice with asthma and Yin deficiency syndrome. *Zhongguo Zhongyao Zazhi* 42, 758–762. <https://doi.org/10.19540/j.cnki.cjcm.20170121.037>.
- Miao, W.G., Tang, C., Ye, Y., Quinn, R.J., Feng, Y., 2019. Traditional Chinese medicine extraction method by ethanol delivers drug-like molecules. *Chin. J. Nat. Med.* 17, 713–720. [https://doi.org/10.1016/s1875-5364\(19\)30086-x](https://doi.org/10.1016/s1875-5364(19)30086-x).
- Nair, M.S., Huang, Y., Fidock, D.A., Polyak, S.J., Wagener, J., Towler, M.J., Weathers, P. J., 2021. Artemisia annua L. extracts inhibit the in vitro replication of SARS-CoV-2 and two of its variants. *J. Ethnopharmacol.* 274, 114016. <https://doi.org/10.1016/j.jep.2021.114016>.
- Parvez, M.K., Rishi, V., 2019. Herb-drug interactions and hepatotoxicity. *Curr. Drug Metabol.* 20, 275–282. <https://doi.org/10.2174/1389200220666190325141422>.
- Qi, H., Liu, R., Zheng, W., Zhang, L., Ungvari, G.S., Ng, C.H., Wang, G., Xiang, Y.T., 2020. Efficacy and safety of traditional Chinese medicine for Tourette's syndrome: a meta-analysis of randomized controlled trials. *Asian J Psychiatr* 47, 101853. <https://doi.org/10.1016/j.ajp.2019.101853>.
- Song, J., Lee, D., Min, B., Bae, J.S., Chang, G.T., Kim, H., 2017. Safety evaluation of Astragalus extract mixture HT042 and its constituent herbs in Sprague-Dawley rats. *Phytomedicine* 32, 59–67. <https://doi.org/10.1016/j.phymed.2017.03.005>.
- Su, X., Zhu, Z.H., Zhang, L., Wang, Q., Xu, M.M., Lu, C., Zhu, Y., Zeng, J., Duan, J.A., Zhao, M., 2021. Anti-inflammatory property and functional substances of Lonicera Japonica Caulis. *J. Ethnopharmacol.* 267, 113502. <https://doi.org/10.1016/j.jep.2020.113502>.
- Wang, Y., Wang, H., Zhang, L., Zhang, Y., Sheng, Y., Deng, G., Li, S., Cao, N., Guan, H., Cheng, X., Wang, C., 2019. Subchronic toxicity and concomitant toxicokinetics of long-term oral administration of total alkaloid extracts from seeds of Peganum harmala Linn: a 28-day study in rats. *J. Ethnopharmacol.* 238, 111866. <https://doi.org/10.1016/j.jep.2019.111866>.
- Wang, Y., Zhang, H., Jiang, J.M., Zheng, D., Tan, H.S., Tang, L.M., Xu, H.X., 2019b. Multiorgan toxicity induced by EtOH extract of fructus psoraleae in Wistar rats. *Phytomedicine* 58, 152874. <https://doi.org/10.1016/j.phymed.2019.152874>.
- Wang, Z.W., Fu, X.Y., Yao, N., Gong, Z.H., Cheng, X.L., Duan, H.J., Ren, Y., 2018. Effects of combined administration of Radix Angelicae Sinensis and hydrocortisone on the therapeutic action in the murine asthma model. *Zhongguo Ying Yong Sheng Li Xue Za Zhi* 34, 379–384. <https://doi.org/10.12047/j.cjap.5648.2018.087>.
- Yang, L., Hou, A., Wang, S., Zhang, J., Man, W., Guo, X., Yang, B., Jiang, H., Kuang, H., Wang, Q., 2020. A review of the Botany, traditional use, phytochemistry, analytical methods, pharmacological effects, and toxicity of angelicae pubescens radix. *Evid Based Complement Alternat Med.* 2020 7460781. <https://doi.org/10.1155/2020/7460781>.
- Yuen, C.W., Murugaiyah, V., Najimudin, N., Azzam, G., 2021. Danshen (Salvia miltiorrhiza) water extract shows potential neuroprotective effects in Caenorhabditis elegans. *J. Ethnopharmacol.* 266, 113418. <https://doi.org/10.1016/j.jep.2020.113418>.
- Yun, J.W., Che, J.H., Kwon, E., Kim, Y.S., Kim, S.H., You, J.R., Kim, W.H., Kim, H.H., Kang, B.C., 2015. Safety evaluation of Angelica gigas: genotoxicity and 13-weeks oral subchronic toxicity in rats. *Regul. Toxicol. Pharmacol.* 72, 473–480. <https://doi.org/10.1016/j.yrtph.2015.05.025>.
- Zhao, L., Li, P., Xu, H., Han, B., Chen, J., Gao, Z., Li, J., Li, X., Wu, C., 2020. Toxicological safety evaluation in acute and 28-day studies of aqueous extract from Bei-Qi-Wu-Jia formula. *J. Ethnopharmacol.* 248, 112324. <https://doi.org/10.1016/j.jep.2019.112324>.